Job Application Auto-Mailer & Lead Scraper

Version 2.0 - Complete Setup Manual & Operational Guide

Two-Phase Autonomous Pipeline: Lead Scraping (Agent Reach Engine) + AI Outreach

1. Overview & System Architecture

The Job Application Auto-Mailer is an agentic, two-phase command-line application built to streamline cold outreach. It eliminates manual job prospecting by scraping recruitment emails across target regions and automatically composing hyper-personalized job inquiries with your resume attached.

- Phase 1 (Lead Discovery): Leverages web scraping techniques from Agent Reach (including Jina Reader r.jina.ai and search crawlers) to identify companies in any specified city and extract verified recruitment/contact emails.
- Phase 2 (Safe Outreach): Utilizes Google Gemini AI to write custom pitches tailored to each company's mission, attaches your resume PDF, and dispatches via your personal email account with anti-spam cooldowns (20-45s) and duplicate logging.

2. Step-by-Step Configuration Guide

Step 2.1 - Gmail 16-Character App Password (Crucial)

Google requires a dedicated App Password for SMTP automation to protect your main account. Never use your standard Google account login password:

1. Open your browser and go to your Google Account Security settings.
2. Ensure '2-Step Verification' is turned ON.
3. Navigate directly to: https://myaccount.google.com/apppasswords
4. In the app name box, type 'JobMailer' and click 'Create'.
5. Google will display a 16-character code (e.g., 'abcd efgh ijkl mnop'). Copy this code.

```
SENDER_PASSWORD="abcd efgh ijkl mnop"
```

Step 2.2 - Free Google Gemini API Key

Gemini AI personalizes your email subject lines and body copy based on company notes:
1. Visit Google AI Studio at: https://aistudio.google.com/
2. Click 'Get API key' and generate a new free key.
3. Copy the key and paste it into your .env file:

```
GEMINI_API_KEY="AIzaSyYourGeneratedGeminiKeyHere"
```

Step 2.3 - Profile & Credentials Configuration (.env)

Edit the '.env' file in your project folder ('job_application_mailer/.env') with your personal details:

```
SENDER_NAME="Hamza Zaidi"
SENDER_EMAIL="hamzazaidi253@gmail.com"
SENDER_PHONE="+212 626640792"
SENDER_LINKEDIN="www.linkedin.com/in/hamza-zaidi-b2a6b7351"
CANDIDATE_SUMMARY="Software Engineer specializing in Python, React, and cloud architectures."
SMTP_SERVER="smtp.gmail.com"
SMTP_PORT=587
RESUME_FILENAME="resume.pdf"
```

Step 2.4 - Placing Your Resume PDF

Place your CV / resume inside the 'data/resumes/' folder. Make sure the filename on disk matches the 'RESUME_FILENAME' value defined in your '.env' file.

```
C:\Users\Pc\Desktop\job_application_mailer\data\resumes\resume.pdf
```

3. How to Use the Application (CLI)

Starting the Interactive Menu

Open PowerShell or Command Prompt, navigate to the folder, and run:

```
cd C:\Users\Pc\Desktop\job_application_mailer
python main.py
```

This opens the interactive menu with all core capabilities:

```
Select an action:
  1. [Search] Scrape & Find Emails in an Area (Phase 1 -> Phase 2)
  2. [Preview] AI-Generated Email (Dry-Run)
  3. [Send] Live Batch Outreach with Resume Attachment
  4. [Inspect] View CSV & Pending Companies
  5. [Test] SMTP Connection & Resume Setup
  6. Exit
```

Phase 1: Automated Scraping by Area (Option 1)

When you choose Option 1, the app prompts you for:
- Target City / Area (e.g., 'Rabat', 'Casablanca', 'Paris', 'Dubai')
- Industry / Keywords (e.g., 'Startups IT', 'Agences Web', 'EdTech')
- Target Job Role (e.g., 'Développeur Full Stack', 'Software Engineer')
- Maximum leads to find (e.g., 10)

The scraper searches the web, probes career pages, extracts verified contact/recruitment emails, and writes them directly to 'data/companies.csv' (automatically skipping duplicates).

Phase 2: Outreach Execution (Option 2 & 3)

- Preview (Dry-Run): View the exact AI-written email, subject line, and attached resume path without sending.
- Live Sending: Dispatches emails to uncontacted companies. Requires typing 'yes' to confirm, applies a randomized 20-45s delay between sends, and records each result in 'data/sent_log.csv'.

4. Direct Command-Line Flags (Power Users)

You can bypass the interactive menu by using CLI flags directly:

```
# Scrape companies in any area:
python main.py --scrape --area "Casablanca" --industry "startups" --role "Développeur Web" --limit 10

# Test credentials and resume:
python main.py --test

# View pending vs sent statistics:
python main.py --status

# Preview generated emails without sending:
python main.py --preview 2

# Start live sending batch:
python main.py --send
```

5. Deliverability & Anti-Spam Safety Rules

[TIP] Anti-Spam Safety Guidelines
Keep daily sends between 30-50 emails. The built-in 20-45s jitter cooldown prevents Google spam flags.

1. Never spam blast: Sending 200+ emails in 5 minutes will lock your account. Keep batch sizes sensible (30-50/day).
2. Built-in cooldown: The random jitter delay (20 to 45 seconds) mimics human behavior and protects sender reputation.
3. Zero duplicate guarantee: 'sent_log.csv' automatically tracks every recipient. Running the app repeatedly will NEVER send a duplicate email to the same company.